

BrightLight Data Analytics

SQL Exercise: Aggregate Functions & Grouping

Instructions:

1. Write on paper with a pen.
2. Draw tables of the final outcome.
3. In the **SELECT** statement, you can select the columns of your choice.
4. After completing this task, scan it to a PDF file and email it to:
rofhiwa@brightlighttutorials.co.za
5. **Submission Deadline:** Friday, March 14, 2025, 00:00

Database: employees_db

Assume you have a table called **employees** with the following structure:

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. COUNT() Function

Write a SQL query to find the **total number of employees** in the company.

SELECT COUNT(*) AS total_employees
FROM Employees;

2. SUM() Function

Write a SQL query to find the **total salary** paid to all employees in the **IT department**.

```
SELECT SUM(salary) As total-salary;  
FROM Employees, where department = 'IT'
```

3. AVG() Function

Write a SQL query to calculate the **average salary** of employees in the **HR department**.

```
SELECT AVG(salary) As average-salary FROM Emplo-  
Where department = 'HR department'
```

4. MIN() and MAX() Functions

Write a SQL query to find the **highest and lowest salary** in the company.

```
SELECT MAX(salary) As highest-salary  
Min(salary) As lowest-salary; FROM Employees
```

5. GROUP BY Statement

Write a SQL query to **group employees by department** and display the **total salary** paid in each department.

```
SELECT department, SUM(salary) As total-department-salary  
FROM Employees; Group by department
```

6. GROUP BY and COUNT()

Write a SQL query to count how many employees work in each **city**.

```
SELECT Count(*) employees; Group by City
```

7. GROUP BY and ORDER BY

Write a SQL query to group employees by **department**, calculate the **average salary** in each department, and **order the results in descending order** of average salary.

```
SELECT department, AVG(salary) As average-salary; From employees  
Group by department; Order by average-salary DESC
```

8. HAVING Clause

Write a SQL query to find **departments where the total salary paid exceeds 100,000**.

(Use GROUP BY and HAVING)

```
SELECT department, SUM(salary) As total-salary; From Employees  
Group By department; HAVING salary > 100,000
```

9. Combining GROUP BY, HAVING, and ORDER BY

Write a SQL query to list **cities where more than one employee works**, ordered by the number of employees in descending order.

```
SELECT cities, SUM(employee) As total-employees;  
FROM Employees
```

10. Combining Aggregate Functions

```
Group by City  
Having > 1  
Order by employees DESC
```

Write a SQL query to find the **department with the highest average salary**.

```
SELECT department,  
FROM employees,  
HAVING AVG (MAX) salary;
```